## API Response Analysis - E-Commerce API (DummyJSON) - Revised

*Tool: Postman | Environment: DummyJSON - Test | Base URL: https://dummyjson.com | Result column is formula-driven (Expected vs Actual status)*

| Test Case No | Test Scenario | Method | Endpoint | Auth Needed? | Required Request Fields | Expected Status | Actual Status | Response Time (ms) | Key Response Fields | Error Message | Result | BA Observation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| TC-01 | Login with valid credentials | POST | /auth/login | No | username, password (JSON body) | 200 | 200 | 1,540 | accessToken, refreshToken, id, username | None | Pass | Token returned here is reused in TC-02 through the {{token}} environment variable. |
| TC-02 | Get current user with valid token | GET | /auth/me | Yes | Authorization: Bearer token | 200 | 200 | 1,130 | id, username, email, firstName, lastName | None | Pass | Protected endpoint; works only with a valid Bearer token (see TC-09 and TC-10). |
| TC-03 | Retrieve the list of all products | GET | /products | No | None | 200 | 200 | 2,900 | products, total, skip, limit | None | Pass | Slowest response in the run with large nested data. Test the limit and skip query parameters to support a pagination requirement. |
| TC-04 | Retrieve a single product by ID | GET | /products/1 | No | Product ID in URL | 200 | 200 | 528 | id, title, price, category, stock | None | Pass | Fastest successful read; returns full detail for one product, suitable for a product detail page. |
| TC-05 | Add a new product | POST | /products/add | No | title, price (JSON body) | 201 | 201 | 1,030 | id, title, price | None | Pass | Returns 201 Created. Verify persistence: call GET on the returned ID afterwards and record whether the product exists (public demo APIs often simulate writes). |
| TC-06 | Update an existing product | PUT | /products/1 | No | Product ID in URL; fields to change (e.g. title) | 200 | 200 | 1,260 | id, title, price, stock | None | Pass | Returns 200 with updated values. Verify with GET whether the change persists. |
| TC-07 | Delete an existing product | DELETE | /products/1 | No | Product ID in URL | 200 | 200 | 1,200 | id, title, price | None | Pass | Returns 200 with the product data. Check for any delete-confirmation fields in the response and whether GET /products/1 still returns the product afterwards. |
| TC-08 | Login with incorrect password | POST | /auth/login | No | username, wrong password (JSON body) | 400 | 400 | 950 | message | Invalid Credentials | Pass | Clear, user-friendly message; 400 is used for bad credentials. |
| TC-09 | Get current user without a token | GET | /auth/me | Yes | None (no token sent) | 401 | 401 | 1,100 | message | Access Token Is Required | Pass | Distinct message for a missing token (compare TC-10). |
| TC-10 | Get current user with an invalid token | GET | /auth/me | Yes | Authorization: Bearer (invalid token) | 401 | 401 | 2,410 | message | Invalid/Expired Token | Pass | Separates an invalid token from a missing one (TC-09), which helps the client decide between re-login and retry. |
| TC-11 | Retrieve a non-existent product | GET | /products/99999 | No | Product ID in URL (non-existent) | 404 | 404 | 376 | message | Product with ID 99999 not found | Pass | Correct 404 for a missing resource; the message includes the requested ID. |
| TC-12 | Add a product with an empty request body | POST | /products/add | No | Empty JSON body {} | 400 | | | To be recorded after re-run | To be recorded after re-run | Pending re-run | Original test put {} in the URL (404, route not found), so it did not test validation. Re-run as POST /products/add with body {}. Expected 400 is a BA assumption: the API should reject missing required fields. |

## Summary and Notes

| Metric | Value |
|---|---|
| Total test cases | 12 |
| Passed (expected = actual) | 11 |
| Gaps (expected differs from actual) | 0 |
| Pending re-run | 1 |
| Average response time (ms) | 1,311 |
| Slowest response (ms) | 2,900 |
| Fastest response (ms) | 376 |

| Actual status code | Count |
|---|---|
| 200 | 6 |
| 201 | 1 |
| 400 | 1 |
| 401 | 2 |
| 404 | 1 |

**Notes and assumptions**

1. Response times were converted from the values recorded during testing (seconds to ms) and come from a single run on a public API, so they are indicative only.

2. TC-12 is pending: the original test used /products/add/{} (the {} was in the URL). Re-run as POST /products/add with body {} and update Actual Status, Response Time, Key Response Fields and Error Message.

3. The expected status of 400 for TC-12 is a BA assumption (required fields should be validated), not documented API behaviour.

4. Key response fields are the actual field names that matter for requirements; verify them against your own Postman responses before finalising.

5. Result is calculated by formula: Pass if Expected equals Actual, Gap if they differ, Pending re-run if Actual is blank.